

EXPLORATORY DATA ANALYSIS REPORT

E-Commerce Order Dataset

DecodeLabs Data Analytics Internship

Project 2

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Executive Summary

This report presents the Exploratory Data Analysis (EDA) conducted for Project 2 of the DecodeLabs Data Analytics Internship, focused on an e-commerce order dataset covering customer orders, product mix, pricing, payment methods, and order fulfillment status.

The analysis combined descriptive statistics, outlier detection, and pivot-table-based trend analysis in Excel to answer four core business questions: how pricing and order sizes behave, how order volumes evolve over time, whether coupon codes influence spending, and how order outcomes and payment methods are distributed. A second layer of analysis translated these findings into revenue and fulfillment KPIs.

The headline finding is a large gap between potential and realized revenue: of a total expected revenue of \$1.26M, only \$242.6K (19.18%) was actually realized, driven by a low delivery success rate of 19.25% and high volumes of cancelled, returned, and pending orders. The report closes with ten operational and marketing recommendations aimed at closing this gap.

1. Introduction

This report covers Project 2 of the DecodeLabs Data Analytics Internship an Exploratory Data Analysis (EDA) on an e-commerce order dataset containing customer orders, products, pricing, payment method, and order status.

2. Problem Statement

The goal of this analysis was to:

- Understand pricing patterns across products and orders
- Analyze order trends over time (monthly and yearly)
- Determine whether coupon codes affect customer spending
- Examine how order outcomes and payment methods are distributed
- Assess whether outliers in the data are errors or genuine customer behaviors

...and ultimately turn these findings into insights the business can act on.

3. Dataset Overview

The dataset has 1,200 order records across 14 columns, grouped by type before analysis:

Column Type	Columns
Numeric	Quantity, UnitPrice, ItemsInCart, TotalPrice
Categorical	Product, PaymentMethod, OrderStatus, ReferralSource, CouponCode

Time / Trend	Date
Identifier (not used in stats)	OrderID, CustomerID, TrackingNumber, ShippingAddress

OrderID, CustomerID, and TrackingNumber were left out of the statistics since they're unique per row. ShippingAddress was mostly placeholder data. PaymentMethod included Cash alongside digital options, consistent with a Cash on Delivery setup.

4. Methodology

Columns were sorted by type, then descriptive statistics (mean, median, standard deviation, skewness, min, max, count) were computed on all numeric columns in Excel.

- Outliers were checked using the IQR method and boxplots
- Pivot tables were used to study monthly and yearly order trends (2023, 2024, and available 2025 months)
- Categories such as OrderStatus, PaymentMethod, and CouponCode were compared against order counts and average order value
- Specific months (e.g. June 2024) were broken down further by product, coupon usage, and referral source

5. Descriptive Statistics

Full descriptive statistics were computed in Excel for the four numeric variables (Quantity, UnitPrice, ItemsInCart, TotalPrice) over all 1,200 orders:

Statistic	Quantity	UnitPrice	ItemsInCart	TotalPrice
Mean	2.9458	356.41	5.485	1,053.97
Standard Error	0.0406	5.692	0.0659	23.667
Median	3	364.21	5	823.62
Mode	1	127.18	5	635.90
Standard Deviation	1.4076	197.18	2.282	819.86
Sample Variance	1.9812	38,878.83	5.207	672,164.78
Kurtosis	-1.2946	-1.1910	-0.7086	-0.0404
Skewness	0.0279	-0.0265	0.0009	0.8914
Range	4	688.54	9	3,445.01

Minimum	1	11.39	1	11.39
Maximum	5	699.93	10	3,456.40
Sum	3,535	427,695.30	6,582	1,264,761.96
Count	1,200	1,200	1,200	1,200

Interpretation

- Quantity, UnitPrice, and ItemsInCart all have skewness close to zero (0.028, -0.027, and 0.0009 respectively) and negative kurtosis, confirming they are close to symmetric, slightly “flatter-than-normal” distributions with no heavy tails consistent with balanced, evenly-spread purchasing behavior.
- TotalPrice is the exception: its skewness of 0.89 confirms a positive (right) skew a long tail of high-value orders pulls the mean (\$1,053.97) well above the median (\$823.62). This is expected, since TotalPrice compounds Quantity and UnitPrice rather than being independently distributed.
- Standard deviation relative to the mean is largest for TotalPrice ($\approx 78\%$ of the mean) and UnitPrice ($\approx 55\%$ of the mean), reflecting real product-price diversity in the catalog, while Quantity and ItemsInCart vary far less (consistent with typical basket sizes).

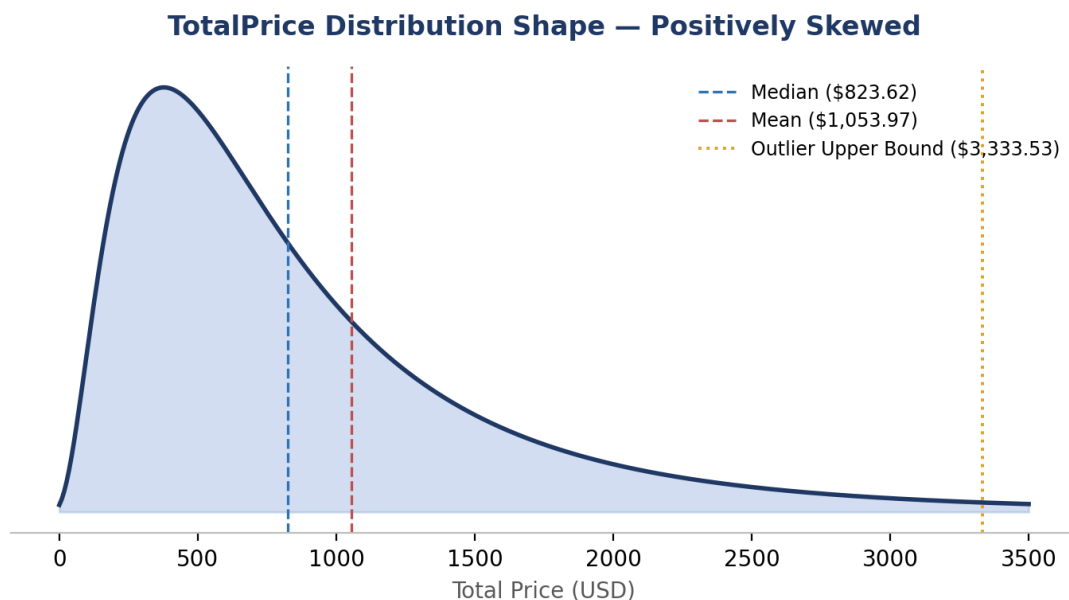


Figure 1 — TotalPrice distribution shape (illustrative curve derived from the reported mean, median, and skewness), showing the right-skew and the outlier boundary.

6. Outlier Analysis (IQR Method)

Outlier boundaries were calculated for each numeric variable using the $1.5 \times \text{IQR}$ rule (Lower Bound = $Q1 - 1.5 \times \text{IQR}$, Upper Bound = $Q3 + 1.5 \times \text{IQR}$):

Variable	Q1	Q3	IQR	Lower Bound	Upper Bound	Max Observed	Outliers Flagged
Quantity	2	4	2	-1	7	5	None
UnitPrice	185.63	521.91	336.28	-318.80	1,026.33	699.93	None
ItemsInCart	4	7	3	-1	12	10	None
TotalPrice	410.52	1,579.73	1,169.21	-1,343.29	3,333.53	3,456.40	8 orders

Interpretation

- For Quantity, UnitPrice, and ItemsInCart, the maximum observed value in the dataset stays below the calculated upper bound, so none of these variables contain statistical outliers reinforcing that order sizes and prices are well-behaved.
- TotalPrice is the only variable where the maximum observed value (3,456.40) exceeds the upper bound (3,333.53). This flags 8 orders as statistical outliers the same 8 identified in the Findings section, all combining a Quantity of 5 with a high-cost product (Tablet, Monitor, Laptop, or Printer). Because they trace back to a clear, consistent cause rather than random noise, they are treated as genuine bulk orders and kept in the analysis rather than removed as data errors.

7. Findings

- Prices and order sizes are fairly balanced. UnitPrice, Quantity, and ItemsInCart are all close to symmetric, with no unusual skew.
- TotalPrice is skewed, but the outliers are real, not errors. 8 orders were flagged, all with Quantity at 5 and high-cost products (Tablet, Monitor, Laptop, Printer) genuine bulk orders, not mistakes.
- Revenue dropped from 2023 to 2024 mainly because of fewer orders, not smaller ones. Average order value barely changed ($\approx 1,084$ vs $\approx 1,046$), but order count fell from 510 to 459.
- There's a recurring mid-year sales peak, but the timing shifts. 2023 peaked in May, 2024 in June, and 2025 (through June) is trending the same way.
- The June 2024 spike wasn't about coupons. NO COUPON orders were actually high that month too. Phone and Printer were the top sellers, pointing to organic demand.
- Cancelled orders have the highest average value (1,106); Returned orders the lowest (985). The orders most likely to fall through are some of the more valuable ones.
- Phone underperforms across the board lowest order count, lowest units sold, lowest revenue, and lowest average order value of any product.
- Order status and payment method are evenly spread, with no single group dominating.

9. Correlation analysis showed that Quantity and Unit Price are strongly related (0.79) customers who buy more expensive products also tend to buy more units of them, rather than buying less because the price is higher. This suggests price alone isn't discouraging bulk purchases, which is useful to know before making any pricing decisions.

8. Key Findings — Business Impact Summary

Beyond the descriptive statistics, the dataset was translated into revenue and fulfillment KPIs to assess overall business performance:

1. The business generated a Total Expected Revenue of \$1.26M, but only \$242.6K was realized from successfully delivered orders, indicating a significant gap between expected and actual revenue.
2. Out of 1,200 total orders, only 231 orders were delivered, resulting in a low Delivery Success Rate of 19.25%.
3. The Revenue Realization Rate was 19.18%, showing that a large portion of potential revenue was lost due to cancelled, returned, pending, or undelivered orders.
4. The business recorded high operational inefficiencies, with 250 Cancelled Orders, 247 Returned Orders, and 237 Pending Orders.
5. Customers added a total of 6,582 items to cart, with an average of 5.49 items per order, indicating strong customer interest and engagement.
6. Despite high cart activity, the Cart Conversion Rate was 53.71%, suggesting that nearly half of carted items were not converted into completed purchases.
7. The average income per delivered order was \$1,050.22, indicating that successfully completed orders generated relatively high revenue value.
8. Laptop was identified as the most delivered product, making it the highest-performing product category in terms of successful fulfillment.
9. Instagram emerged as the most frequently used referral source, indicating strong customer acquisition through social media traffic.
10. However, Email generated the highest actual revenue, suggesting that email marketing attracted more valuable or higher-converting customers.
11. The coupon code "FREESHIP" recorded the highest usage among customers, highlighting the strong influence of shipping incentives on purchasing decisions.
12. The year 2024 generated the highest revenue, indicating stronger sales performance during that period.

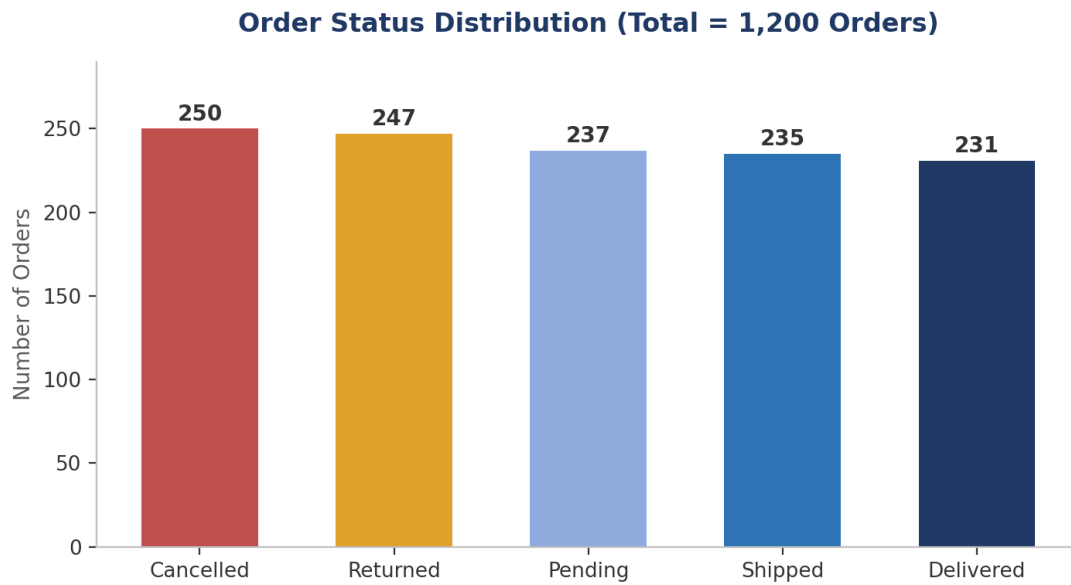


Figure 2 — Order status distribution across all 1,200 orders. Only 231 orders (19.25%) were delivered.

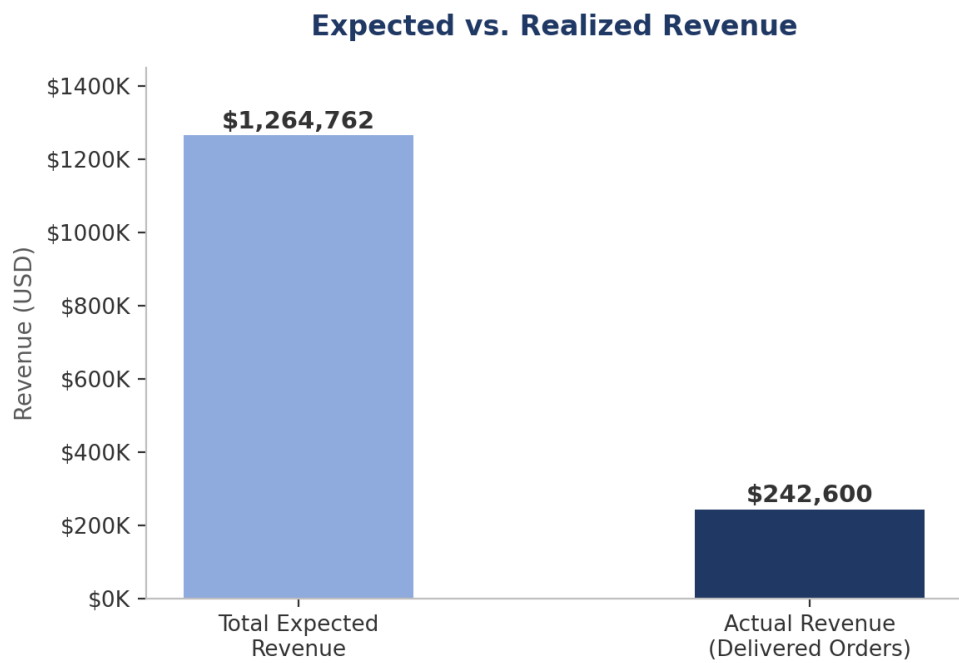


Figure 3 — Expected revenue (\$1.26M) vs. revenue actually realized from delivered orders (\$242.6K) — a Revenue Realization Rate of 19.18%.

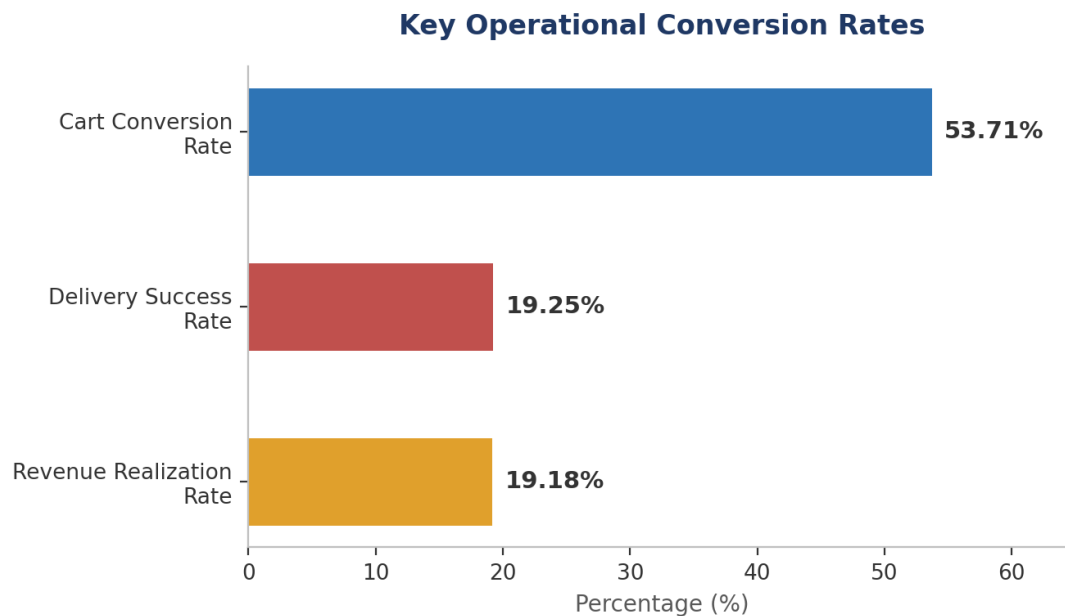


Figure 4 — Key operational conversion rates: Cart Conversion, Delivery Success, and Revenue Realization.

9. Recommendations

1. Improve order fulfillment and logistics processes to reduce the high number of cancelled, returned, and pending orders.
2. Investigate the root causes of failed deliveries and returned products to improve customer satisfaction and revenue realization.
3. Strengthen inventory management and order tracking systems to minimize operational delays and incomplete deliveries.
4. Increase investment in Email Marketing campaigns, as it generated the highest actual revenue despite Instagram driving more traffic.
5. Continue leveraging Instagram for customer acquisition, while optimizing conversion strategies to turn traffic into completed purchases.
6. Introduce cart abandonment strategies such as reminder emails, targeted discounts, and limited-time offers to improve the cart conversion rate.
7. Expand promotion and availability of high-performing products such as Laptops, since they contributed significantly to successful deliveries.
8. Continue offering shipping-related incentives like the FREESHIP coupon code, as customers responded positively to shipping discounts.
9. Monitor delivery performance KPIs regularly Delivery Success Rate, Revenue Realization Rate, and Cart Conversion Rate to track operational improvements over time.

10. Implement customer feedback collection for returned and cancelled orders to identify recurring service or product issues.

10. Conclusion

This exploratory analysis shows that the e-commerce dataset's underlying pricing and purchasing behavior are healthy and predictable prices and order sizes are balanced, bulk-order outliers are genuine, and pricing does not appear to discourage larger purchases. The primary weakness is operational: only about one in five orders is actually delivered, and this fulfillment gap not weak demand is what separates \$1.26M in potential revenue from the \$242.6K actually realized.

Closing this gap depends less on generating more traffic or interest (cart activity and referral traffic are both strong) and more on fixing delivery reliability, reducing cancellations and returns, and converting the marketing channels that already work email in particular into a larger share of completed orders. The recommendations above provide a starting roadmap for that operational improvement.